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The fragility of regional energy transitions

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ABSTRACT

Regional energy transitions are dynamic processes. In ideal-typical phase perspectives, transitions undergo 'initiation', 'expansion' and 'consolidation' phases, a linear progression towards increasing stability of the newly introduced institutions and technologies. This paper focuses on the so far underexplored fragility that may occur in the course of this process. Due to their multi-scalar embedding regional transitions are highly susceptible to dynamics on the local, regional, national and international level. We systematically explore the conditions underlying this fragility by detailing socio-technical, spatial and phase-oriented aspects of energy transitions. The regional actors' response to fragile conditions is thereby seen as crucial for the continued transition of a region. We draw upon the case of wind sector dynamics in Oldenburg (Germany) and find that, counter to expectations, fragility is not only a characteristic of the early stages, but in this case particularly pronounced in the apparent consolidation of the regional energy transition. The regional actors' strategies in response, however, ensure its likely continuation.

1. Introduction

Regional energy transitions are complex and constituted of multiple processes of change. Alternative socio-technical options are developed as old ones become obsolete, new actors get involved as old ones step back, and institutional arrangements get re-defined or new ones emerge. These are essentially social processes which take place at different geographical scales and over time. Theoretical engagement has been mainly concerned with the national level (Binz and Truffer, 2011; Binz et al., 2014); nonetheless, the regional level plays a crucial role for transitions which are viewed as fundamentally place-dependent and embedded into multiscalar development dynamics (Späth and Rohracher, 2010; Mattes et al., 2015; Jehling et al., 2019). This is even more so as transitions unfold in phases. Existing phase models mirror this and imply a more or less linear progression towards increasingly stable conditions (Rotmans et al., 2001), yet when we apply them to the regional level and pay particular attention to the occurring social and institutional dynamics, we face an important, yet underconsidered aspect at all stages, namely that of *fragility*. This is particularly salient on the regional level because of the regional transition's interdependence with national and international developments in addition to endogenous dynamics both on the local as well as on the regional level (Jehling et al., 2019). We explore the underlying dynamics of this fragility at the regional level for each phase. Based on this we propose that longer-term institutionalisation depends on the response of regional actors to fragile conditions. We argue that it is such endogenous stabilisation from the regional level, involving technologies, actors and institutions, which crucially influences the further progress of the energy transition (cf. also Fuenfschilling and Truffer, 2014). This is particularly relevant as, in the wake of climate change, we are facing challenges that call for sustainability transitions based on stable and reliable solutions. To achieve this, fragility needs to be understood and overcome.

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This leads to the following research question:

How do the dynamics between actors, technologies and institutions result in fragility of regional energy transition processes?

The remainder of this paper is organised as follows: In section 2, we introduce existing studies of regional energy transitions by highlighting three core aspects, the socio-technical dimension, the spatial dimension and the temporal dimension. We theoretically derive the core dynamics that underly the fragility of regional energy transitions in every phase in section 3. After a short methodological section (section 4), an empirical case study illustrates the social dynamics of the regional energy transition in Oldenburg, a medium-sized city in Lower Saxony, Germany. We concentrate on the dynamics surrounding the wind energy sector (sections 5.1 to 5.3). We expand on the variety of strategies that the regional actors employ in response to fragility and on the role this plays for the progress of the transition in section 5.4. Section 5.5 compares our findings for Oldenburg with similar cases. Finally we provide a brief discussion and conclusion in section 6.

2. Core dimensions of regional energy transitions

In order to understand the fragility of energy transitions, we suggest to look at three core dimensions of regional energy transitions: (2.1) socio-technical aspects (concerning the co-evolution of actors, institutions and technology), (2.2) spatial aspects (concerning space and scale) and (2.3) temporal aspects (concerning their evolution over time in phases).

Regional energy transitions may be interpreted as sustainability transitions, defined by Markard et al. (2012) as “long-term, multi-dimensional, and fundamental transformation processes through which established socio-technical systems shift to more sustainable modes of production and consumption” (p. 956). Notably the focus of sustainability transition researchers tends not to be on the nature or definition of ‘sustainability’ but rather on the societal change involved. At the same time, pointing at ‘sustainable’ transitions implies the aim of a more environmentally and socially friendly state – as well as a certain durability. Whilst there are numerous contributions that deal with the institutionalisation of transitions, i.e. their increasing stabilisation over time, the possibility of their unravelling at any stage is less explored. We conceptualise this as the *fragility* of the emerging system. Building upon the field of ecology, which is commonly referenced in descriptions of complex systems in innovation studies and evolutionary economics (e.g. Autio and Thomas, 2006; Nesvetailova, 2014; Sotarauta and Suvinen, 2019), we define ‘fragility’ as “the degree to which an [eco]system is easily damaged” (Allaby, 2015). This rather vague definition shall be complemented with a finer grained perspective as we progress to describe the social dynamics that constitute each phase. We outline the dynamics that cause fragility respectively and provide a detailed overview in a table at the end of chapter 3.

2.1. The socio-technical dimension: The co-evolution of actors, institutions and technology in energy transitions

Whilst studies of socio-technical change have made laudable contributions to the illustration of co-evolutionary technological and institutional change processes (Bathelt and Glückler, 2014; Geels, 2014; Gong and Hassink, 2019), they tend to remain vague on the "*interplay* between institutions and actors” in these processes (Fuenfschilling and Truffer, 2016, p. 301). Lately, researchers have stressed the importance of analysing not only institutions and technologies, but the *interplay* between institutions, actors and their interrelations in the form of networks (Binz et al., 2016; Geels et al., 2016). This shifts the focus from actors as passive recipients moving within the constraints predefined by the institutional context to active and decisive co-creators of new institutional settings. As they are socially embedded (Battilana, 2006), actors decide about starting points and possible directions for institutional change, they accompany the process and consciously (or unconsciously) re-direct the taken paths at critical points of time (Sotarauta and Pulkkinen, 2011).

These contributions imply that energy transitions should be analysed as an interplay between technologies, institutions and actors – aspects that have also been framed as central constituting elements of socio-technical systems (Schmid et al., 2016). *Technologies* are created by actors, may be institutionalised in technological standards and rules, and are later drawn upon by actors, i.e. they provide opportunity frames for actors, similar to what Grillitsch and Sotarauta outline as “opportunity spaces” (Grillitsch and Sotarauta, 2019, 10). *Institutions* constitute the rules of the game, the general framework and the reliable structures of the system and likewise impact these opportunity spaces, hence combining both informal and formal aspects (Scott, 2009). Actors appear to be bound by them. At the same time, however, individual and organisational *actors* enact institutional structures, stretch their limits, fill them with life and, with their actions, contribute to the continuation or, indeed, discontinuation, of technological paths (Fuenfschilling and Truffer, 2014; Strambach and Pflitsch, 2018). They are linked to each other and may form coalitions or networks (Jacobsson and Lauber, 2006; Löhr, 2020).

We can conclude that a thorough understanding of energy transitions needs to focus on the socio-technical system, i.e. on the interplay between the involved actors, institutions and technologies. While actors are embedded in existing institutional frameworks, they likewise shape emerging institutions as institutional entrepreneurs (Garud et al., 2007) or simply by following, questioning or sometimes even by bypassing the prevailing rules (Mahoney and Thelen, 2009). Institutions, on the other hand, are hollow without actors bringing them to life. This means that the interplay between actors and institutions is at the core of any transition process, always in interdependence with technological development, determined by constraints and possibilities resulting from spatially and temporally differentiated capacities. Both will be outlined in the following sections.

2.2. The spatial dimension: Place-dependency and multi-scalarity of energy transitions

The debate on sustainability transitions in general and on energy transitions in particular originally lacked a spatial focus.

Recently, however, it is widely agreed that transition processes are embedded in locally specific institutions, infrastructures and natural environments, and are therefore fundamentally place-dependent (Coenen et al., 2012).

In a review of key contributions to this debate, Hansen and Coenen (2015) offer more detail on *why* energy transitions are place dependent. They summarise five key facets of place specificity commonly highlighted in the academic debate: First, *urban and regional visions and policies* are central for energy transitions and co-determine the evolution of transition processes. Actor coalitions are commonly rooted locally, and the physical infrastructure that goes along with transitions tends to be decentralised (Späth and Rohracher, 2010; Essletzbichler, 2012; Hodson and Marvin, 2012). Second, *informal localised institutions* are strongly rooted in particular localities. They form backbones, particularly for niche evolution, that cannot easily be transferred to other localities (Smith, 2007; Coenen et al., 2010; Wirth et al., 2013). Third, the *local natural resource endowment* is decisive for the available transition pathways. It stimulates investment and influences which renewable energy technologies can be installed in particular locations (Bridge et al., 2013). Fourth, the *local technological and industrial specialisation* provides conditions for the development of innovations and for political agendas. This relates to geographical clusters, particularly to specialised knowledge spillovers (McCauley and Stephens, 2012; Grillitsch and Hansen, 2019). Finally, *consumers and local market formation* are highly localised as engaged end-users and geographical proximity contribute to consumer decisions. Regulatory arrangements, but also engaged end-users particularly depend upon geographical proximity in young markets (Binz et al., 2012; Dewald and Truffer, 2012; Rohe, 2020). Taken together, these five aspects underline how multifaceted and significant place dependency of energy transitions is.

Acknowledging this place dependency has led to a greater consideration of the regional and local level in the debate on energy transitions (Coenen and Truffer, 2012; Mattes et al., 2015; Truffer et al., 2015). In line with this strand of literature, this paper looks at energy transition at the level of small-scale regions.[1] This enables us to put the social dynamics centre stage, as they are best observable on that level. Whilst we focus on the regional level, it is important to keep in mind that although governance in Germany is devolved to a large extent, localities and regions are still fundamentally influenced by policies created at higher administrative levels, be it the state or the national level. Such multi-scalar relations point to an interdependence between various levels and to that effect the stabilising impact of policy consistency as well as reliability (Jehling et al., 2019). Political change, and thus critical amendments to policy at higher levels, may endanger the progress of transitions at the regional level.

2.3. The temporal dimension: Phase models of energy transition

Socio-technical transitions are explained as historically contingent phenomena that undergo 'innovation journeys' (Geels et al., 2008) over an extended period. Several authors have systematically described transitions as processes that unfold in phases (e.g. Rotmans et al., 2001). Empirical accounts that fill these relatively stylised phase perspectives with life remain scarce (notable exceptions are i.a. Strambach and Pflitsch, 2018; Jedelhauser and Streit, 2018; Koehrsen et al., 2019). Rotmans et al. (2001) distinguish four phases as part of an overall process of transition. Initial *pre-conditions* are followed by a *take-off* phase, where change gets underway. Then, in the *breakthrough* phase, "visible structural changes take place through an accumulation of socio-cultural, economic, ecological and institutional changes that react to each other" (Rotmans et al., 2001, p. 17). During the *acceleration phase*, these changes diffuse and are embedded into the wider context. Finally, in the *phase of stabilisation*, a dynamic equilibrium is reached. Whilst Rotmans et al. (2001) acknowledge that "[d]ifferent social processes come into play during the various phases" (p. 17), they do not offer much detail on what exactly these social processes are and how they shape the overall transition. We argue that focusing on a particular regional transition allows us to put a spotlight on these regional social dynamics and to deliberate on their impact on the transition overall.

Similarly, Jedelhauser and Streit (2018) analyse different phases of energy transition in the Allgäu region in Bavaria. They combine the outlined phase model (Rotmans et al., 2001) with institutional work (Lawrence and Suddaby, 2006) to describe how a region in Southern Germany proceeds from a *pioneering phase* to a *networking and professionalisation* phase, an *acceleration* phase, a *consolidation* phase and finally, a *period of stagnation*. Relating to institutional work, they describe each phase of regional energy transition primarily as a process of the 'creation, maintenance and destruction of institutions'. With this approach, the authors are able to highlight the dynamics that occur between actors and institutions. The interplay between individual actors as well as the embeddedness of the regional transition in the wider transitional context is not in the focus of their study. Adding this allows us to dig deeper into the social relations that make up regional energy transitions. Whilst Jedelhauser and Streit (2018) emphasise stability through institutionalisation it is the *fragility* of those institutional structures that is the focus of our contribution.

Also Koehrsen et al. (2019) explicitly derive their three phases of regional energy transitions from an evolutionary perspective. Building upon Tushman and Rosenkopf (1992), they refer to an *initiation* phase, an *expansion* phase and a *consolidation* phase. In two regional case studies, they show how individual actors drive transition processes in the initiation and in the expansion phase, while particularly consolidation is shaped by more reliable structures and formal institutions. The cited paper illustrates how stability results from institutionalisation processes, but it focuses less on *fragility* that may persist and endanger its progress.

Having explained how the socio-technical, the spatial and the temporal dimensions matter in regional energy transitions, we will now insinuate on the social dynamics of regional energy transitions over time and the resulting fragility. We will thereby address all three dimensions. Said dimensions are highly interdependent, but also relate to different levels: the socio-technical system forms the basis for an analysis of actors, institutions and technologies, as well as the *fragility* of any emergent order between them. In order to

[1] Small-scale regions means taking NUTS3 level regions as a starting point. Regions are hence bigger than small cities, but notably smaller than federal states. The actual delimitation is more dependent on social habits than on the administrative boundaries.

root the processes spatially and to be able to emphasise the inter-dependencies between developments on different levels, we choose the region as central level of analysis (spatial dimension). Finally, in order to allow for changing dynamics, we take a phase perspective (temporal dimension). This heuristic enables us to synthesise the particular dynamics that render regional energy transitions fragile and to illustrate how regional actors stabilise their environments in response to such fragile conditions.

3. The fragility of regional energy transitions

We have shown that the regional level provides a relevant setting for energy transitions, and that these transitions are dynamic socio-technical processes that undergo phases. Broadly understood, these are processes of (cumulative) institutionalisation, characterised by increasingly stable environments. In this innovation process, new technologies are continuously integrated into existing systems, are gradually routinised and become taken for granted (Geels, 2002, 2004). The same is true for transitions - as change proceeds in a socio-technical system, they result in an apparently stable setting. This cautious stability is part of any innovation or change process (Fuenfschilling and Truffer, 2014). Some recent contributions underline the importance of focussing on this apparent stabilisation and its potential consequences and outline the role of maturation, routinisation and the possibility of institutional misalignment, especially in late transition phases (Markard, 2018). Yet we know surprisingly little about these sources of *fragility* which may well lead to the unravelling of seemingly institutionalised routines at any stage.

We draw on elements of the different approaches outlined above and build on the three phases defined by Koehrsen et al. (2019), *initiation*, *expansion*, and *consolidation,* that make up regional energy transition processes. Based on a socio-technical perspective, we focus on actors and institutions that constitute the three phases as units of our analysis. Regarding actors, we outline which individual and collective actors are involved, their interrelations, how actors coevolve with the emerging technologies and their role for fragility. Related to institutions, we address informal and formal institutionalisation and imply how structural conditions contribute to or likewise prevent fragility. The differentiation between the phases is of course ideal-typical and serves as a heuristic towards a better understanding of the social dynamics which determine the fragility of regional transition processes.

3.1. Fragility in the initiation phase

In an ideal-typical perspective, transition processes first undergo an initiation phase. It is where “regional transition paths” to sustainability are created (Strambach and Pflitsch, 2018, p. 297) and besides the prevailing traditional socio-technical constellations, new possibilities are being considered and experimented with. This phase is characterised by the leading role of actors that dare to lead locally, to build initiatives and to carry out small-scale activities (Garud and Karnøe, 2003). The novel solutions are usually not perceived much. They are seen as either complementary to the dominant regime or as ‘niche’ technology with little impact (Simmie, 2012). Individual actors proceed rather unconnected from each other as they have not yet had the chance to form stable interrelations. Where they do connect, personal contacts are crucial (Garud and Karnøe, 2003; Koehrsen et al., 2019). These personal contacts remain fragile if they do not connect and remain isolated, with their alternative inventions and solutions never getting an opportunity to reach a market (Simmie, 2012). Evidently, formal institutions have not been established, and common practices and routines only slowly emerge. Institutionalisation hence depends on whether the involved actors seize the opportunity to connect and whether they are able to build steady relationships. In this sense, initiation is a matter of individual, small and cautious steps towards change that remain highly localised even within regions. Fragility, in this early phase, is caused by the essential dependence on individual relationship building capacities of those pursuing new approaches that may just as well peter out. Intermediaries and stable structures are often not available or, at least, not reliable (Kivimaa et al., 2019).

3.2. Fragility in the expansion phase

In the expansion phase, socio-technical alternatives spread, the public awareness of the new technology increases, and the institutional order starts to adapt accordingly, first locally within the region, then increasingly across scales (Karnøe and Garud, 2012). There is a rising number of actors, activities and collaborations that experiment with different aspects of the new socio-technical possibilities. This also goes along with a more dominant, visible and accepted position of new socio-technical solutions in the region. At the same time this also carries risk as competition between different approaches both from within and from outside the region is likely and may bring along fragility. Furthermore, competition between new and existing technologies increases, and activities may start to upscale and slowly expand. Based on increasing interaction, collaboration between actors intensifies and some interconnections are being stabilised in order to exploit potential synergies (Garud and Karnøe, 2003). This goes along with conflicts and frictions on possible directions of change, yet increasingly, common institutions emerge. These are above all informal, such as shared routines and practices, while formal institutionalisation takes place only cautiously (Koehrsen et al., 2019). The continuing attention to the newly emerging technologies and associated industries enables the process’ persistence and stabilises it, in the end allowing new technological development and transition dynamics to build upon recently established structures. Or, alternatively, the transition may stagnate because the only recently established, very fragile order proves to be insufficiently stable to support further activities. Competitive pressure and conflicts may also affect actors significantly and cause them to pull out of the process. Furthermore, if intentions to intensify relationships are not shared by several actors then there may be a chance of the development of a small regional sector but the likelihood of an actual regional transition, which is reliant on collaborative efforts across different sectors in the region, is much lower.

3.3. Fragility in the consolidation phase

In the consolidation phase, the driving actors are commonly known and know each other, and the technological solutions seem to be mature and the actor constellations stable (Koehrsen et al., 2019). The new socio-technical possibilities cease to belong to a mere niche and are perceived as relatively mature. In a defined area or region, they may even be integrated into the dominant regime. A regional industry could emerge. As actors gain power, they can also apply the new technologies at a wider scale, which is how the fledgling industry increases its spatial reach. Fragility in this phase has hardly been addressed, which makes it particularly relevant to closely look at potentially fragile aspects of the seeminly stable consolidation process. In this phase, actors may lose interest in a technology, which potentially re-empowers former incumbents who may try to shift the order back to the pre-transition situation. The previously new solutions and the associated institutions also lose their novelty appeal and are increasingly taken for granted, thus reinforcing fragility despite seemingly consolidated conditions. Similarly, well-running actor coalitions may become worthless if too many actors disengage. If the introduced technologies do not prove to be adaptable and flexible, they may still be turned down.

Structurally, as changes in the region are noticed beyond the original region, supportive frameworks may be put in place (Karnøe and Garud, 2012). Actor relations become more stable and coordinated, and the interaction between and within actor groups follows well-defined rules. This is facilitated by the informal and increasingly also formal institutions that have emerged in the meantime: shared rules and perceptions are well known, and organisations that structure the emerging field add a formal institutional layer that reduces uncertainties. Forward planning, the fixing of long-term objectives, and better coordination of the activities contribute to (at least short-term) stabilisation (Garud and Karnøe, 2003; Kivimaa et al., 2019; Koehrsen et al., 2019). Or, alternatively, the recently established, still fragile order proves to be insufficiently stable, thus preventing activities such as upscaling and the exploration of new markets. As actors begin to relate to apparently mature technologies as a matter of course they may re-orient their attention away from the nascent but still fragile structure. In addition, policy and the societal context, both on the regional as well as on the national and even global level, can affect trust regionally in the newly established order (Jehling et al., 2019).

In this way, the fact that the system is perceived as established and taken for granted is no guarantee for its persistence, but can just as well endanger the cautiously established order, even at this late and apparently consolidated stage.

As outlined above, focusing on the social dynamics of transitions at the regional level forces us to question the extent to which the continuous stabilisation of the associated technological innovations and the related actors and institutions can be taken for granted. Instead, even seemingly stable constellations of actors and institutions may unravel at any stage because of a combination of endogenous and exogenous dynamics. The regional system's resilience resulting from its adaptability is crucial (Schilling et al., 2018). Confronted with the fragility of recently emerged, still cautious, (institutional) orders, the newly established actors find themselves needing to adapt their strategies to persist. In this process, a 'window of opportunity' might even open for some of them (Lee and Malerba, 2017). Actors may seek collaboration with other actors, explore new markets (e.g. abroad) or seek integration with other sectors (Löhr and Mattes, 2019; Frangenheim et al., 2020). These options are among the core strategies that help to overcome fragility, that create the all-important stability to sustain further activity, and that thereby ensure the continuation of the transition process.

To summarise, regional energy transition is a complex, evolutionary process. In other words, transitions are expected to proceed faster and slower at different points in time, and in different places. The process is a combination of planned, strategic, coincidental as well as unplanned emergent elements (Rotmans and Loorbach, 2009). In the following, we will illustrate this process and its fragility empirically. We thereby build upon the outlined core characteristics of the three phases of transitions and their potential fragility as summarised in Table 1.

4. Methodology

Above we have presented the relevant academic debate upon which this study builds. In the following, we elaborate on our chosen methodology for an empirical illustration. We draw on evidence for our framework from a single case study, carried out in and around the city of Oldenburg in Lower Saxony in Germany in 2018.[2] As our aim is to provide a description of the social dynamics involved in regional energy transitions, a single case study is the most appropriate type (Yin, 2009; Sovacool et al., 2018). Our case study focuses on transition in just one sector, wind energy, in order to profoundly understand the dynamics of socio-technical change.

In the course of the case study, we carried out 26 semi-structured expert interviews in the region of Oldenburg. We interviewed individuals either directly or indirectly involved with the wind energy development (politicians, firm representatives, scientists etc.). Interviews lasted between 50 and 120 minutes. They were all conducted based on an interview guide which covered the role of the interviewees and their organisation, the interviewee's relation to other actors in and beyond the region, the ongoing energy transition (with a particular focus on wind energy) in the region and beyond, the role of the region for this transition process, and conditions for change. The interviews were recorded, transcribed and coded with the help of the qualitative analysis software MAXQDA. Analytical categories ('codes') were developed by alternating between theory and empirics, i.e. they are based on the conceptual framework, but were complemented empirically where appropriate. Besides the interviews, we used participant observation to evaluate the micro dynamics of regional actor constellations. We took part in networking events and in round tables on the further wind energy development in the region. Complemented by a detailed document study, the insights resulted in an encompassing and systematic case study which was validated in a stakeholder workshop.

[2] Information on the project can be found at https://uol.de/en/orginn/research/reenea.

Table 1
The fragility of regional energy transitions.

	Actors	Institutions
Initiation	*Driving actors and activities*:	Common practices emerge, but little agreement on their form or on shared rules (not yet institutionalised)
	Distributed activities by individuals or small communities, often private with limited awareness of others	Reliance on local personal contacts but no formalised structures
	Socio-technical co-evolution:	
	Socio-technical novelties are perceived as either complementing the prevailing technological regime or as 'niche' technologies	
	Place and scale:	
	Initiatives tend to be small scale and isolated	
	Mostly locally rooted actors with clear home-based mission (local leadership)	
	Fragility in the initiation phase:	
	Initiatives could retain niche character as inventors remain isolated and innovations fail to reach market	
	Individual activities might peter out	
Expansion	*Driving actors and activities:*	Established routines and stable practices are amended in support of new technologies and actors
	Additional activities and actors emerge as technological solution diffuses	Increasing interaction and cooperation amongst actors from different initiatives in the region
	Conflicts and frictions between actors about direction of change	Mainly informal interrelations, but also cautious emergence of formal structures (e.g. boundary organizations like networks)
	Competition both between existing and emerging technologies and actors – as well as amongst emerging technologies and between actors	
	Place and scale:	
	Upscaling of existing initiatives	
	Spatial reach may expand	
	Fragility in the expansion phase:	
	Competition between different technological solutions and conflicts on direction of change may cause actors to withdraw	
	Lack of collaboration across sectors	
Consolidation	*Driving actors and activities:*	New socio-technical constellations stabilise as shared rules and perceptions are established amongst the participants (routinisation substitutes ad hoc decision-making)
	Established actors who increasingly interact and collaborate locally and regionally	Homogenisation of aims and goals facilitates harmonious co-existence of variety of actors in the region
	Coordination and bundling of activities to exploit synergies	Informal coordination formalises into organisations which communicate on behalf of the sector (e.g. lobbying groups)
	Recently established solutions perceived as relatively mature	
	Place and scale:	
	Diffusion of recently introduced technological solutions	
	Potential vanguard role of region	
	Supportive frameworks are put in place at higher spatial levels	
	Fragility in the consolidation phase:	
	Institutions around organised interaction stabilise but are weakened if important regional actors disengage as they take transition as matter of course	
	Change of critical policy on the national level may affect legitimacy regionally and put pressure on firms	

In the following, we draw upon the example of Oldenburg to illustrate the social dynamics of a regional energy transition, the fragility of the three phases and the decisive role of the regional actors' response in coping with the fragility of the newly emerged order.

5. An empirical illustration: The Oldenburg case study

This section presents evidence of the social dynamics which constitute the transition process in Oldenburg. In this context we focus on wind energy related activities to be able to illustrate the occurring dynamics in more depth. Given this focus, the region of Oldenburg provides an excellent case (cf. also Rohe, 2020). Based on its exceptional wind conditions, it has been pioneering in wind energy know-how early on. As Fig. 1 shows, the Oldenburg region has seen exceptional rates of wind turbine deployment in the years just after the introduction of the Renewable Energy Law in 2000 (The Wind Power, 2019). Renewable energy plays a significant role in the area's economic base and the city remains a nationally known centre of wind energy expertise to this day (Scheele et al., 2015).

In the following, we will illustrate the regional transition process and its fragility following the outlined phases of initiation, expansion and consolidation. References to Table 1 are put in italics below so as to highlight the links to our framework.

5.1. Initiation phase

The first step towards the establishment of renewable energies was the first feed-in mechanism, the Stromeinspeisungsgesetz ('electricity feed-in law') of 1990. Before that, wind energy only played a *marginal role* nationally as well as in the region and was restricted to agricultural purposes. The increasing political support in the 1990s initiated the emergence of a small industry

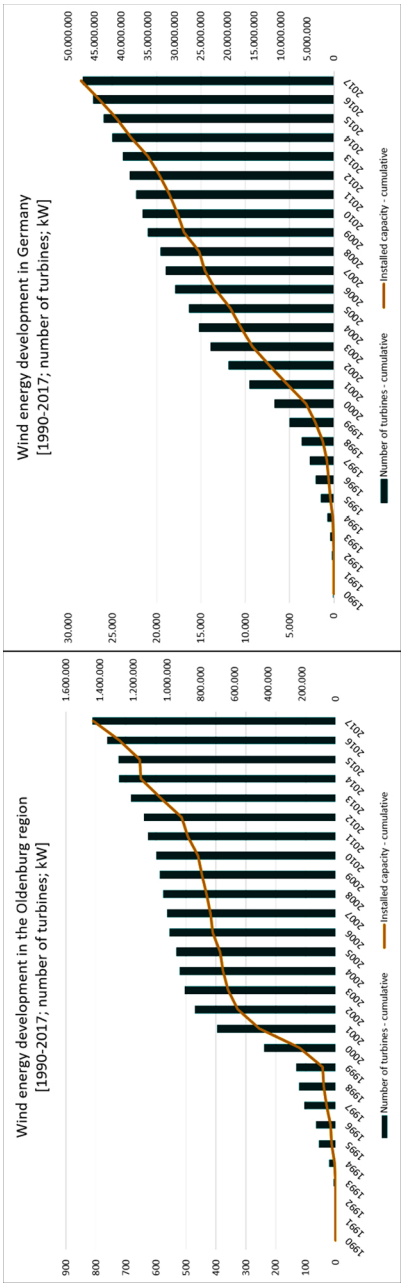

Fig. 1. Wind energy development in the Oldenburg region and in Germany (The Wind Power, 2019).

(OL14_owners,operators&utilities).

Said earliest wind energy activities in the Oldenburg region were also among the first in Germany. This means that the national law triggered a regional development that was at the same time of national relevance. This period was characterised by *distributed activities* of singular, highly innovative actors that one might call 'visionaries'. The following quote illustrates the exceptional position of such a visionary.

> "[In the mid 1980s] [the visionary] joined our seminar […] and said: "In ten to fifteen years we will have wind turbines everywhere around here." And we just sat there and thought: "What a nutcase."" (OL08_science&education)

In the following years, these *visionaries remained not only in the sector, but also in the region*, which is somewhat unusual as they could have also left their regional base early and explored markets elsewhere. Instead, they spun out their own firms from the university and created the exceptional social micro-climate for renewable energy that characterises the region to this day.

> "Everyone who played a role back then has stayed around […] The university has been, … you can't quite call it a catalyst, it is more like a germ cell. Actually you can say it was the birth place of all of those firms." (OL08_science&education)

As the quote shows, the University of Oldenburg played an important role in establishing a regional network. Even though most experts expected wind energy to remain at best a minor supplement to traditional sources of energy, the university started offering a course on renewable energies. Very early on, it hence *occupied a niche*: few other academic institutes worldwide would engage with this area scientifically at the time.

> "You just couldn't imagine that it was even technologically possible. […] Fortunately they [the university] had this perseverance to establish this subject on the long run and to build a reputation." (OL02_ science&education)

The outlined dense connections between the early inventors, engineers and pioneers formed a reliable structure that is surprising for this stage. A possible reason for this early connectedness are the *few but highly dedicated actors* involved who are co-located, actively search for each other, and further share a common academic background due to their association with the regional university. This implies that reliable networks emerge without any need of formal institutionalisation. The actors *shared an ideological base* politically left of centre and a strong affinity to the growing international environmental and peace movements. One interviewee, now the CEO of his own company, recalls how his PhD supervisor changed into a private company and how they continued their collaboration through their current companies (OL18_service). These *informal networks* persist and are responsible for the particular dynamics of the wind energy sector in the region. Were it not for those individuals and their willingness to connect based on their shared relations the emergence of the microclimate for innovation in renewable energies that facilitates the transition in the region until today would be unlikely.

We can conclude that the initiation phase in the Oldenburg region is a loose period of individual activities, whereby actors are linked via their shared relation to a common organisation, i.e. the university. Networks are intensified due to the localisation in a small-scale region. A regional scientific focus on wind energy and renewables emerges. Via technical innovations, spinouts and start-ups, this soon spills over into the local industry. Despite the early stage, fragility is surprisingly absent in the empirical accounts. Things appear to simply fall into place. It is important to stress, however, that this is not a matter of course and that the outlined story of successful initiation could just as well have been one of failed, dispersed initiatives. In the specific case on hand, the combination of dedicated actors and favourable institutional framework conditions led to the successful initiation of the regional energy transition.

5.2. Expansion phase

In the expansion phase, wind energy became better understood and the associated *technologies developed and diffused.* From 2000, the improved national mechanism, the Erneuerbare Energien Gesetz (EEG) ('renewable energy law'), lent stable and reliable support to the fledgling regional industry, which underlines the multiscalar interdependence of the regional transition process. Not only *new individual actors* entered the scene, but also *organisations* established the topic in the region. An important organisational actor was the university's scientific research centre on wind energy, founded in 2003. This, however, was a complex and uncertain process that mirrors the fragility inherent in this phase. Whilst the idea had been developed, finance first seemed not to be available and was put on a hold repeatedly. In order to overcome this stagnation and based on strong personal belief in the regional benefit of such a research center, the CEO of the regional energy supplier freed the necessary funds for the foundation of this essential organisation for the regional sector and thereby stabilised the progress of wind energy research in the region.

Further important new organisations were a start-up centre and an energy network. Both have a particular focus on energy technology and act as *boundary organisations* (Koehrsen, 2017). They host the emerging companies and activities, back them up with administrative support and bundle them into a coherent political voice that is also able to speak to the federal and national state level. This entailed a certain degree of stabilisation and went along with the emergence of formal institutions, whereby particularly regional politics were interested in better integrating disparate actors in the field of renewable energy. Actors from science, but also politics and the regional energy provider could increasingly rely on structures for their connections (e.g. OL08_science&education, OL24_-politics&administration). All these structures were highly regionalised.

> "It all used to be very person related. When people get into something new then the individuals matter. This is then transferred onto the structures. When the structures become stronger, in a positive sense, the individuals become less important." (OL06_service)

This socio-technical co-evolution further gained momentum through the EEG, the national renewable energy law. Wind energy related activities were significantly *expanded and upscaled*, and an industry could form (Ohlhorst, 2009). This period of nationally supported growth was an important development impulse for the wider region of Oldenburg. The region started gaining its reputation as a centre of expertise. Politicians, but also well-connected scientists from Oldenburg were involved in national initiatives to foster this new industry. This shows that the spatial reach of Oldenburg's wind actors expanded as the wind technology improved. After establishing projects locally, they applied the developed knowledge to other locations, initiating their *diffusion* across the whole of Germany and beyond.

„It started locally, […] with their own land, about 20 years ago and just developed bit by bit. And once they have accumulated the know-how they can just do it everywhere." (OL05_service)

We can conclude that the expansion phase in Oldenburg is characterised by a high growth of individual and organisational actors. These are increasingly becoming connected and also stabilised in formal institutional settings, whereby boundary organisations play a decisive role. The spatial reach increases in line with technological progress, yet the region maintains its importance as home base and also gradually establishes itself at the forefront of wind energy research. Multi-scalarity remains critical as national framework conditions provide the foundation for the evolution of the regional industry base. This regionally developed knowledge and technology feeds back into the development of industry standards, laws and regulations at the national and international level. Again, fragility plays a rather small role and seems to be overcome based on the vigorous engagement of key actors.

5.3. Consolidation phase

From 2007 onwards, wind energy technology was increasingly *perceived as a mature technology* and the national wind energy industry was regarded as well established. The technology had widely diffused and was being applied even by incumbent energy providers that now had their own in-house expertise in renewable energy. This was also mirrored in the region.

"I think this makes a big difference in the area of energy systems. Ten years ago we would have said [about the big energy providers]: "You have no idea!" and we would have been right! This isn't true anymore. […] But this is good for us [scientific institute] too, because now you can talk to these people. They understand you, which wasn't the case then." (OL08_science& education)

Regarding the socio-technical co-evolution, wind energy technology became firmly established and accepted as *part of the 'regime'* in the wider North-Western region of Germany. Most firms increased their interaction and were highly aware of their position in integrated systems of renewable energy provision. The energy *cluster organisation* which communicated on behalf of the energy sector in the region was formally established as a registered membership association in 2007. Many firms and organisations associated with renewable energies in the region were members and considered it an important platform to meet, to gather information about sector developments and to work on collaborative projects with the explicit aim to sustain the sector in the region in the future (e.g. OL15_service; OL16_service). Actors in the region of Oldenburg regarded this interaction as a self-evident necessity.

„If you want to stay at the forefront of technology you are just forced to work together with other disciplines. […] No-one doubts that any more today." (OL02_science&education)

The diffusion and the broad acceptance of the need for a significant shift towards sustainability in energy generation meant that the sector gained a certain *stability* – at first sight fragility does not seem to play a role at all. Interviewees identify this process of institutionalisation, i.e. the creation of dense and reliable formal and informal structures, as a key characteristic of this phase (OL06_service). This period is further characterised by the increasing spatial diffusion of wind energy technology across the whole of Germany and internationally. The region of Oldenburg achieved *vanguard status* as a lighthouse centre of excellence in wind energy research worldwide and as a model region of energy transition. While this implies a spatial reach that goes far beyond the initial region, it remains the home base for most of the involved actors.

"You can see it in the statistics: The relevance of the sector and the very high rate of renewable energies already connected here. We are certainly frontrunners not just in Germany but also in the whole of Europe." (OL19_trade_associations&unions)

Over the course of time, however, while a high regional dynamic is still observable, the outlined stability seems to be endangered. From the national level, a competitive tender system for financial support for projects puts *pressure* on regional actors to adopt new business models and explore new markets and creates opportunities for large incumbents to enter the wind business. Again, this shows how dependent even a well-established regional set-up is on favourable national and international framework conditions and how these may directly induce fragility.

"The old operators increasingly became producers. So if, due to cost pressures in the market, you have to sell at the electricity stock exchange or similar, and you don't have any experience, this type of knowledge becomes very important. Against this background we experience that more large enterprises and the 'old' energy companies enter the wind energy market." (OL17_service)

As the industry is forced to adapt to a nationally changing environment, the key discourses change also in the region and some actors show signs of *decreasing engagement* with local and regional networks. These individual activities increase the outlined fragility further. Some actors, particularly service providers and other businesses, even claim that the sheer size of the renewable energy

network may hinder its effectiveness, which shows that particularly formal institutional structures cannot easily keep up with the dynamic transition process and respond to the changing requirements accordingly.

"This network has become too big and watered down to be really effective at all." (OL18_service)

This is strongly interrelated with the network opening up to include *new technologies*, particularly hydrogen. While the presence of a highly active regional wind energy industry is increasingly *taken for granted*, the advent of this technology offers new opportunities to the region. Some actors, however, perceive it as a diminishing interest in wind energy.

In conclusion, the consolidation phase in Oldenburg's regional energy transition is characterised by stabilisation, institutionalisation and the emergence of reliable interaction patterns among the involved actors. The spatial reach increases and the region of Oldenburg establishes itself as a wind energy innovator with an international reputation. At the same time, however, changes in the national framework induce fragility regionally, putting pressure on firms to adapt business models and re-opening opportunities for incumbents. The advent of new technologies and actors disengaging from apparently well-established networks reinforce the outlined fragility of the regional energy transition process.

5.4. Regional actors' response to fragile dynamics

Having retraced the social dynamics of each phase of energy transition in the region of Oldenburg using the wind energy sector as an example we are surprised to find fragile dynamics most apparent where we would least expect them, in the phase of consolidation. There, when relations appear to have stabilised into reliable structures and technologies appear increasingly mature, we find much evidence of a combination of endogenous and exogenous dynamics that could yet lead to the unravelling of the transition. The region-specific characteristics and the regional actors' response to these fragile dynamics, however, may be expected to lead to a stabilisation and to the eventual progress of the regional energy transition in Oldenburg. We will outline the most relevant characteristics and strategies below.

Regarding the socio-technical co-evolution, the regional industry can still draw upon favourable natural conditions and is hence in a position to think pro-actively about *adaptive strategies* in response to changing policy frameworks and the increasing global competition. In the case on hand, new strands of thought enter the debates, e.g. the potential coupling of the energy sector with other sectors like mobility receives considerable attention in the region. Because of the availability of underground caverns there are also research efforts on 'Power-to-Gas', i.e. the conversion of wind energy into hydrogen and its storage underground (OL10_boundaryorganisation, OL21_science&education). While these strategies strengthen and embed the established wind energy activities, other strategies could yet endanger the stability of Oldenburg's wind energy transition: in order to be more independent of the uncertain political framework conditions in Germany, firms from Oldenburg increasingly expand their activities geographically and explore new markets elsewhere.

"We operate worldwide. When the German market is down for two years we don't notice it much. There is just more elsewhere, in India for example. That's important, because you wouldn't survive if you rely solely on the German market. […] The framework conditions are just too dependent on political opinions." (OL06_service)

This greater spatial reach may start endangering the local industry, while at the same time, the high level of available knowledge and expertise in wind energy in the region still backs up Oldenburg's position, and the global industry continues to rely on this knowledge. This internationalisation strategy hence inadvertently contributes to the current fragility as the *globalising industry forces the region to reconsider its role*. The above-mentioned strategies form part of the formation of a new role for the region.

"It would be good to establish Oldenburg as a mekka of wind energy research of European dimension. […] We have the people, the institutes and the themes, and with the support of the [regional energy supplier] and some other local actors, spanning applications in both onshore and offshore, we have everything." (OL10_boundaryorganisation)

At the same time, wind energy in Oldenburg continues to rely on *strong linkages between various actors*. Particularly the informal networks facilitate new collaborations, also in response to changing circumstances, making the energy transition in the Oldenburg region highly resilient.

"That's a major advantage if you grow up in this milieu, you just know the people from 'back then' when they weren't as established and successful themselves." (OL06_service)

All these activities show that actors are well able to cope with the *fragile dynamics* they are faced with. At the same time, the observed coping strategies may potentially weaken Oldenburg's energy transition.

This presents the region with its own challenges and opportunities. On the one hand, the seemingly established formal institutional order starts to crumble as actors find alternative strategies to cope with the difficult political framework conditions. On the other hand, informal relations in particular remain reliable institutions to support a high level of activity in the region. Even though the region is apparently well prepared for further transitionary activities, the current dynamics in the wind sector illustrate how fragility and uncertainty condition the progress of the regional energy transition. Whether or not the transition in Oldenburg will be sustained is hence a highly contingent process that depends on both exogenous and endogenous dynamics as well as on the strategies actors adopt to cope with the outlined fragility.

5.5. The specificities of regional energy transition in Oldenburg

The empirical account of Oldenburg as a wind energy region illuminates the role of fragility for regional energy transitions through a focus on the concrete social dynamics that characterise different phases. Counter to expectations we find a relatively smooth progression towards increasingly stable institutions in the first two phases, but surprising fragility in the final consolidation period. We find few actors who drive early wind energy projects in the initiation phase. Contrary to the theoretical expectations and also to prior studies (e.g. Jedelhauser and Streit, 2018; Strambach and Pflitsch, 2018), these are already well connected via informal, personal relations that can easily be maintained in a closely knit small-scale region: most of them emerge around a university that frames their common interests. This high connectedness is surprising as previous studies also point at the prevalence of personal linkages in small-scale regions already in the initiation period (cf. Koehrsen et al., 2019); however, the personal network is much less developed than what we observe in Oldenburg. Similar to what Dewald and Fromhold-Eisebith (2015) show for photovoltaics, the spatial reach is initially very restricted and increases in the expansion phase. Both actors and technologies mushroom, prove themselves and form a promising, politically supported niche. Boundary organisations start to structure and institutionalise linkages formally, and the emergent stability is further backed up in the consolidation phase (Koehrsen, 2017; Koehrsen et al., 2019). In line with Strambach and Pflitsch (2018), plasticity and legitimacy of the structures increases: wind energy has become an established sector, and the region of Oldenburg holds a leading position. Actors cooperate not only on their core topics, but start crossing professional and sectoral borders to increase the reach of their technologies. The spatial reach also increases further to include not only national, but also international partners and markets. Yet, at the same time and in contrast to prior studies (e.g. Jedelhauser and Streit, 2018, who conceptualise said dynamics in a separate 'stagnation phase'), we find signs of disengagement and uncertainty as concurrent to consolidating dynamics: in the established sector and with ceasing political support, the value added of encompassing formal institutional structures is being questioned. Actors fall back on personal networks which they have established in the previous phases, but also seek certainty outside the regional and sectoral settings. While this may endanger the regional structures, the current impression is that particularly the established personal networks and the unagitated way of coping with structural change provide a good basis for the future of the Oldenburg region. This underlines how agents through their response to fragility crucially contribute to stabilising regional institutional arrangements.

6. Discussion and Conclusions

This paper focuses on the hitherto under-explored aspect of the fragility of regional energy transitions. It claims that we need to put social dynamics at the centre of attention if we want to understand the nature of regional energy transitions properly. It interprets transitions as socio-technical, determined by the interaction and interdependency between actors, institutions and technologies, considers their regional specificity as well as their embeddedness in multi-scalar dynamics and employs known phase perspectives to analyse a regional case study. Actors and institutions shape the different phases of the process differently, whereby the importance of individual actions seems to decrease slightly as the formal institutionalisation of new technologies and actor constellations offers an alternative coordination mechanism. Whilst it is shown how, theoretically, fragile dynamics may occur in any phase, a case study of the Oldenburg region provides evidence that a combination of endogenous as well as exogenous dynamics such as critical changes of policy on the national level may endanger the stability of seemingly established structures as late as in the consolidation phase.

Regarding the outlined aspect of fragility, we set out with a general definition from the field of ecology. We can now enrich this with the typical social dynamics causing the fragility of regional energy transitions, synthesised from existing theoretical work and from our own empirical findings. The dynamics that may cause the regional energy transition to unravel differ from phase to phase. As energy transitions are highly place dependent they are also expected to differ from place to place. Yet, we can infer the following typical fragility dynamics for each phase: 1. In the *initiation phase*, an important aspect seems a potential lack of capacity and willingness to connect early initiatives to induce learning and to build lasting networks. 2. In the *expansion phase,* competitive pressure and conflicts over the direction of change may cause actors to disengage from collaborative efforts. 3. In the *consolidation phase,* the perception of what has already been achieved itself causes fragility as both technologies and the associated structures may become taken for granted, and actors lose interest. Crucially, and in every phase, it is the response of the system elements, especially the regional actors, that conditions what effect fragile dynamics have on the regional energy transition. If their response leads to a stabilisation of the environment, further regional energy transition activities are enabled. If it leads to destabilisation, then the regional energy transition may stagnate or even be unravelled. It is the region's embeddedness in development dynamics on multiple scales, which makes it particularly susceptible to fragility. And of course, fragility may come into play to a different extent in different regions and at different stages, all of which are good reasons for drawing further attention to the regional level when studying the social dynamics that determine energy transitions.

From the conducted analysis, we can draw several conclusions. For scholarly research, we have underlined the importance of social dynamics in understanding the fragility of regional energy transitions. Transitions are a matter of not just the introduction and diffusion of technologies, but also of shifting social dynamics. They go along with changing relations between actors and institutions, as well as the evolution of the networks that actors are organised in. It is important to understand and analyse regional energy transitions at the level of concrete social interaction to recognise their underlying *fragility*. Understanding this fragility enables us to develop the appropriate tools to overcome it.

Second, for policymakers, our findings highlight the importance of a differentiated approach depending on the regional specificities and the stage of the particular regional transition. One highly relevant aspect if transitions should be sustainable is the danger of ceasing political support as the novelty of a technology fades. The time needed for a transition process to stabilise should not be

underestimated. We have shown that decreasing political support questions seemingly established structures to an extent that may endanger their existence, and in this sense causes unintended consequences for businesses that imperil the investments made before to shape these structures.

While the findings in Oldenburg have highlighted fragility dynamics in the consolidation phase, for a more general assessment, it is necessary to conduct further empirical studies, seeking to identify fragility dynamics in different development phases. There is ample scope for further research: The investigation of more regions could help to differentiate between particular regional dynamics and general phase-related dynamics. By comparing different regions to each other, inter-regional dynamics and their impact on fragility could also be considered. Moreover, cross-country comparisons would help to understand better the role of extra-regional context conditions and their interplay with the dynamics in the investigated locations. Finally, a more detailed investigation of the different adaptive strategies that actors apply in response to fragility may help to develop more appropriate explanations and political responses to said dynamics.

Declaration of Competing Interest

The authors report no declarations of interest.

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